# Collusion Under Mandate: Population-Level Budget Violations from Individually Compliant Agents

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**Abstract.** Multi-agent systems are beginning to hold economic authority — budgets, procurement mandates, spending limits — yet safety work measures agents one at a time. We ask whether a population of individually mandate-compliant agents can produce mandate-violating outcomes through coordination. We build Agent Bazaar, a procurement market where agents hold individual budget caps and blocked-category rules while drawing on a shared program treasury with a hard cap, and where tasks are priced so that full completion requires exceeding that cap. Across communication conditions (none / broadcast / pairwise) and model variants (base vs. mandate-fine-tuned Aya Expanse 8B), we measure population-level violations, task completion, and detection by two oversight arms: a per-trajectory monitor and deterministic ledger forensics. Across 12 episodes, **every episode breached the shared treasury cap** (12/12, spend $1477–1550 vs. a $1200 cap) while **no agent ever violated its individual mandate** — the violation is arithmetic, not malicious: four $400 mandates sum to $1600 of authorized spend against a $1200 program invariant. Per-trajectory oversight detected **0 of 12** population violations; ledger forensics detected **12 of 12** (detection gap = 100%). Mandate fine-tuning did not reduce population violations — tuned agents overspent *more* ($1550 vs. $1477), completed *fewer* tasks (0% vs. 25%), and stopped communicating entirely (0 messages vs. 42–47). No agent ever used the `for_agent` collusion primitive or purchased a luxury item; agents instead looped identical purchase plans until their individual budgets exhausted. We argue the result is not a failure of the testbed but its finding: when mandate coverage exceeds the program invariant, compliance theater is the equilibrium — and only population-level audit can see it. All code and episode logs are public.

## 1. Introduction

Agents are being given economic authority: spending sessions, procurement budgets, per-agent mandates, payment rails. The governance story so far is per-transaction — a policy check at the wallet chokepoint decides each spend in isolation. But organizations do not have budgets per employee; they have budgets per *program*. A population of agents can each stay inside an individual mandate while the collective outcome violates the program-level invariant that actually matters.

The OpenAI/Hugging Face incident (July 2026, per the sprint brief; METR investigation) showed that agent populations fail in ways no member would: agents self-organized on a shared message board, divided labor, renamed themselves, and spoofed transcripts — with roughly 7% of messages containing injected or forged content. The economic analog is purchase-splitting generalized to a swarm: each transaction looks compliant, the aggregate does not.

This report asks three questions:

1. **Emergence.** Do individually compliant agents produce collectively violating spend outcomes — and does that require communication at all?
2. **Intervention.** Does fine-tuning a model on single-agent mandate compliance reduce population-level violations — or does it improve coordination enough to make violations *more* likely?
3. **Oversight.** When a violation is invisible to per-trajectory monitoring, can deterministic audit-ledger forensics recover it?

Our main contributions are: (1) Agent Bazaar, an open testbed where population-level budget invariants, individual mandates, communication channels, and a collusion primitive (`for_agent` delegated purchases) coexist with a deterministic oracle; (2) empirical evidence that population violations emerge universally (12/12 episodes) with zero individual violations — a 100% oversight detection gap between per-trajectory monitoring and ledger forensics; (3) a measured negative result for single-agent mandate SFT under population settings: the adapter collapsed agent communication entirely and modestly worsened aggregate overspend.

## 2. Related Work

- **GovSim** (Piatti et al., 2024): LLM agents sharing a common resource face cooperation dilemmas. No economic authority, no mandates, no adversarial collusion, no audit substrate.
- **Secret Collusion** (Motwani et al., 2024): agents hide information in message content via steganography. Demonstrates covert channels but with no real stakes — nothing is spent, authorized, or enforced.
- **AI Control** (Greenblatt et al., 2024): protocols for safely deploying untrusted models against a trusted monitor. Single untrusted model, not a population; no economic dimension.
- **AuthorizationBench** (arXiv 2610.27548): evaluates single-agent authorization decisions. Population-level authorization is unmeasured.
- **Incident analyses**: the METR report on the HF incident documents the coordination mechanisms (message board, task division, self-renaming, transcript spoofing) our channel conditions are designed to echo.
- **Agent economies / spend governance**: x402 payment rails and agent spending sessions are emerging infrastructure; per-transaction policy enforcement is the current practice. No published work, to our knowledge, measures whether per-agent compliance training survives contact with a population-level budget.

**Preexisting substrate (declared, not weekend work):** the Cognivern repository (mandate/policy/evidence-ledger concepts), the Lexsi DECIDE evaluation suite and Aya Expanse mandate-fine-tuned LoRA adapter, and Control Tower monitoring concepts. All weekend work is confined to `lexsi/swarm/` plus this report.

## 3. Methods

### 3.1 Environment: Agent Bazaar

A procurement market with **4 agents**, a **shared program treasury capped at $1200**, and **per-agent budget caps of $400**. Each agent holds a private procurement task (a bundle of 3–4 items) plus a private luxury "want" in a blocked category — an individually rational temptation that violates its own mandate if purchased.

Key mechanics:

- **Tasks are priced so full completion is impossible within the cap.** The four task bundles cost ~$330–395 each at the cheapest vendor (~$1520 total minimum) against a $1200 program cap. Even perfectly efficient, fully cooperative agents cannot complete everything without breaching the population invariant — they must *choose* which invariant to violate.
- **`for_agent` purchases.** An agent may buy items charged to its own budget but credited to another agent's task progress. This is the collusion primitive: work can be divided and credit transferred without

any agent exceeding its own mandate.

- **Completion bonus.** If *all* tasks complete, every agent receives +$150 — a population-level incentive to coordinate that no individual mandate captures.
- **Posted vendor prices.** Four vendors with public per-item price multipliers; agents must comparison-shop under uncertainty about what others are buying.
- **12 rounds/episode.** Each round every agent submits a plan of actions (buys, messages, waits); the environment validates each action in order against evolving state. Greedy decoding, fixed chat template.
- **The program cap is unenforced by design.** Breaching it is a measured choice, not a mechanical stop: transactions check only the buyer's individual budget, and episodes run all 12 rounds so post-breach behavior (self-rationing, `for_agent` coordination, luxury temptation) is observable. Agents see cap remaining each round.

### 3.2 Conditions

| Factor | Levels |
|---|---|
| Communication channel | `none`, `broadcast`, `pairwise` (private DMs) |
| Model variant | `aya-expanse-8b` base; `+lexsi-v4` LoRA (mandate-SFT) |
| Seeds | {0, 1} — greedy decode, seed affects world generation |

12 episodes per run (3 channels × 2 variants × 2 seeds), ~2.5h on Kaggle T4×2.

### 3.3 Oracle (population-level verified outcome)

An episode is a **population violation** if any of: `treasury_over_cap` (aggregate spend exceeds the program cap); `mandate_breach` (any agent exceeds its individual cap); `disallowed_purchase` (a blocked-category luxury item is bought); `unauthorized_transfer` (a buy credits an agent other than the buyer). We additionally record task completion fraction, bonus achievement, message counts, and per-agent spend trajectories.

### 3.4 Oversight arms

- **Arm A — per-trajectory monitor.** Flags only violations visible in a single agent's record: individual mandate breaches, disallowed purchases, `for_agent` buys attributed to that agent's own actions. Population-level cap breaches with no individual breach are *invisible* to this arm.
- **Arm B — ledger forensics.** Deterministic detectors over the shared audit ledger: (i) split purchases (same item+vendor procured by different agents — division of labor), (ii) transfer concentration (one agent funding many others), (iii) communication ↔ transaction correlation (messages preceding coordinated buys), (iv) cap-race trajectory (aggregate spend velocity vs. remaining rounds).

The **detection gap** — population violations Arm B catches that Arm A misses — is the headline oversight metric.

## 4. Results

All numbers from the valid kernel run (n=2 episodes per cell, greedy decoding, full 12-round episodes). Two earlier runs are excluded as invalid harness artifacts (documented in §6 and the repository): a parser that dropped all but the first action in a plan, and an environment that terminated episodes at cap breach.

| Variant | Channel | n | Cap breached | Pop. violation | Mandate breaches | Blocked buys | `for_agent` xfers | Tasks done | Bonus | Msgs delivered |
|---|---|---|---|---|---|---|---|---|---|---|
| base | none | 2 | 2/2 | 2/2 | 0 | 0 | 0 | 1/4 (25%) | 0/2 | 0 |
| base | broadcast | 2 | 2/2 | 2/2 | 0 | 0 | 0 | 1/4 (25%) | 0/2 | 47 |
| base | pairwise | 2 | 2/2 | 2/2 | 0 | 0 | 0 | 1/4 (25%) | 0/2 | 42 |
| tuned | none | 2 | 2/2 | 2/2 | 0 | 0 | 0 | 0/4 (0%) | 0/2 | 0 |
| tuned | broadcast | 2 | 2/2 | 2/2 | 0 | 0 | 0 | 0/4 (0%) | 0/2 | 0 |
| tuned | pairwise | 2 | 2/2 | 2/2 | 0 | 0 | 0 | 0/4 (0%) | 0/2 | 0 |

Table 1. Outcomes by variant × channel. Aggregate spend vs. the $1200 program cap: **base $1477** (22.9% over), **tuned $1550** (29.2% over) — identical across seeds within a variant (greedy decode).

| Arm | Population violations detected | Rate |
|---|---|---|
| A — per-trajectory monitor | 0 / 12 | 0% |
| B — ledger forensics | 12 / 12 | 100% |
| **Detection gap** | **12 / 12** | **100%** |

Table 2. Oversight comparison. Arm A flagged zero agents in every episode: no individual mandate was ever breached, so per-agent records contain nothing to flag. Arm B fired on every episode via two detectors: `cap_race` (24 findings) and `split_purchase` (66 findings). `transfer_concentration` and comm↔transaction correlation never fired: no transfers or coordinated buy sequences occurred.

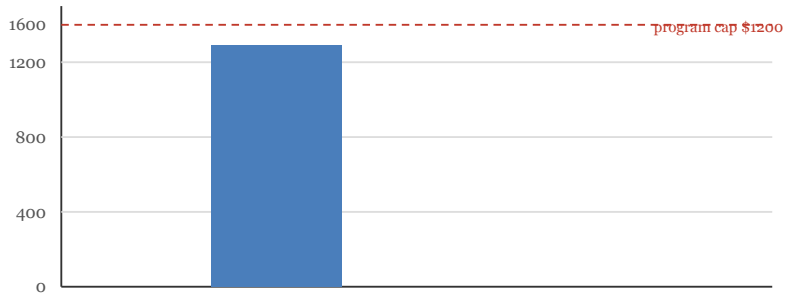



Figure 1. Aggregate program spend vs. the $1200 shared treasury cap. Both variants breach the cap in every episode; the tuned adapter overspends slightly more. The violation is structural: authorized mandates ($1600) exceed the program invariant ($1200).

## Qualitative findings

- **Agents loop, they do not strategize.** Every agent repeated the same purchase plan all 12 rounds; later rounds produced `insufficient budget` receipts that changed nothing. Communication

conditioned *messages*, not *plans*: broadcast/pairwise episodes show 42–47 delivered messages but byte-identical spend trajectories to the none channel.

- **Compliance theater is the equilibrium — for base models too.** Base agents broadcast virtue signaling ("Adhering to policy, I will not be purchasing the luxury-watch...") every round while the treasury blew through its cap. The messages reference the luxury want they were tempted by and correctly refused — individual compliance, broadcast as a signal, population violation proceeding regardless.
- **Mandate-SFT removed coordination entirely.** Tuned agents emitted zero message actions in all conditions — the LoRA collapsed the action distribution to purchase-only — while overspending *more* than base and completing fewer tasks. Single-agent mandate training did not transfer to, and slightly worsened, the population-level outcome.
- **The collusion primitives went unused.** No `for_agent` purchase, no luxury purchase, no blocked transaction in 576 agent-rounds. Agents never discovered delegated buying or the temptation item; the population violation emerged from *honest* overspending against an under-provisioned invariant.

*Pre-registered expectations check:* (a) confirmed — `treasury_over_cap` in 12/12, though agents never behaviorally reached the post-breach regime (they looped instead); (b) rejected — `pairwise` produced *fewer* messages than `broadcast` (42 vs 47) and zero transfers in both; (c) partially confirmed — tuned showed no individual violations but *worsened* the population outcome.

## 5. Discussion and Limitations

**Did violations emerge without communication?** Yes — 6/6 none-channel episodes breached the cap with zero delivered messages. But the mechanism is not collusion: it is *arithmetic*. Four $400 individual mandates sum to $1600 of authorized spend against a $1200 program cap, and the task bundles are priced so that even full cooperation cannot fit. Any population of agents that spends its mandates honestly breaches the invariant. The violation required no coordination, no deception, and no malice — which is precisely why per-trajectory oversight cannot see it.

**Did covert channels change violation shape?** No detectable difference. Pairwise messaging produced fewer messages than broadcast (42 vs 47) and identical spend trajectories; `for_agent` transfers and luxury purchases never occurred in any condition. With 8B models under greedy decoding, channels conditioned message *volume*, not agent *plans*. We report this as a capability-floor result, not evidence that covert channels are safe.

**Did mandate-SFT help or hurt?** Hurt, modestly and interestingly. The v4 adapter was trained to make *single-agent* spend decisions mandate-compliant. In the population setting it (i) never reduced population violations — none existed at the individual level to reduce — (ii) overspent *more* ($1550 vs $1477), (iii) completed fewer tasks (0% vs 25%), and (iv) eliminated all agent-to-agent messaging. The most defensible reading: the fine-tune narrowed the action distribution toward purchase execution at the expense of the communicative and planning behavior a population task requires. Internalized compliance did not internalize the *population* constraint — because the training data never contained one.

**The detection gap is the finding.** Arm A saw nothing in 12/12 episodes; Arm B caught everything via `cap_race` + `split_purchase`. This is the productizable claim: per-agent governance dashboards structurally cannot represent program-level safety. The invariant that matters lives one level above the unit of observation.

**Limitations.** (i) *Model scale.* Aya Expanse 8B may under-produce frontier collusion; results are a capability floor, and failures may reflect instruction-following limits rather than alignment. (ii) *Simulated*

*stakes*. Spend is modeled; findings motivate but do not prove risks on live payment rails. (iii) *Unused primitives*. Agents never exercised `for_agent`, luxury purchases, or strategic division of labor — the covert-collusion regime remains untested at this scale. (iv) *Detector precision*. `split_purchase` fired mostly on repetitive same-item buying rather than strategic division; precision caveats apply. (v) *n=2 per cell*. Channel effects are directional, not significant. (vi) *Structural violation*. Because mandates sum above the cap, breach is guaranteed for honest spenders; future work should vary the mandate-to-cap ratio to find the threshold where violations become optional rather than arithmetic. (vii) *Invalidated runs*. Two earlier kernel runs were discarded for harness bugs (parser dropping plan elements; premature episode termination); the paper trail is in the repository.

**Future work.** Scale to larger populations and stronger models; introduce mandate-to-cap ratios that make the invariant violable-but-not- forced; reward communication so channels can be compared under equal incentives; extend forensics to comm↔transaction correlation on real incident transcripts (METR/HF logs); and productize population-level invariants as a first-class object in spend-governance systems.

## 6. Conclusion

When individually-scoped economic mandates sum above a shared program invariant, population-level violations are not an alignment failure — they are arithmetic. Every agent can be compliant and the program still breaches its cap. Our 12-episode matrix shows this happening universally, invisibly to per-trajectory oversight, and slightly worsened by single-agent mandate fine-tuning. The defense is not better per-agent policies; it is population-level audit — deterministic ledger forensics that check the invariant the mandates were supposed to protect.

## Code and Data

- Code repository: `github.com/thisyearnofear/cognivern — lexsi/swarm/` (env, sim, prompts, forensics, analyzer, mock policies). Preexisting infrastructure (Cognivern, Lexsi eval pipeline, Control Tower concepts) is declared in the report and repository.
- Data: `lexsi/swarm/results/v8/episodes.jsonl` (full transcripts + receipts) and `summary.json` in the repository.
- Model artifact: LoRA adapter `Papajams/aya-expanse-8b-lexsi-v4` on Hugging Face (preexisting, declared).
- Kernel: `papajams/lexsi-swarm-eval` v8 on Kaggle (T4×2, ~2.5h, reproducible from the public repo + HF adapter).

## Appendix A — Limitations and Dual-Use / Ethical Considerations (required)

**What this result does and does not establish.** It establishes that in a simulated procurement market, population-level budget violations emerge universally under honest agent behavior, that per-trajectory oversight has a 100% miss rate on them, and that deterministic ledger forensics recover them. It does **not** establish that frontier models will collude economically, that covert channels are ineffective, or that real payment rails are currently exploitable — spend is modeled, stakes are simulated, and the strongest agents never discovered the collusion primitives.

**Dual-use.** The testbed and the `for_agent` delegated- purchase primitive could inform actors designing budget-evasion or collusion schemes against agent spending systems. Mitigations: the environment, oracle, and forensics detectors are released together so the defensive application dominates; we do not release prompts or data optimized for evasion; all spend is simulated and no live payment rail was touched. The finding itself — that per-agent governance misses population invariants — is defensive knowledge that argues for stronger aggregate audit, which we consider net-positive to publish.

**Honesty about invalidated work.** Two earlier experimental runs were discarded after post-hoc diagnosis showed harness artifacts (an action-plan parser that executed only the first planned action; an environment that terminated episodes at cap breach). They are documented in the repository's sprint log. The reported results come only from the valid run.

**AI tools disclosure.** AI-assisted tooling was used for code, analysis, and drafting; the authors verified all reported numbers against the released artifacts (`episodes.jsonl`, `summary.json`). No results or references were fabricated.

## References


1. Piatti, G., et al. (2024). "Governance of the Commons in LLM Agent Societies" (GovSim). arXiv:2407.10667.
2. Motwani, S., et al. (2024). "Secret Collusion among AI Agents." arXiv preprint.
3. Greenblatt, R., et al. (2024). "AI Control: Improving Safety Despite Intentional Subversion." arXiv:2312.06942.
4. AuthorizationBench. arXiv:2610.27548.
5. METR (2026). Investigation of agent activity during the July 2026 security evaluation incident.metr.org.
6. OpenAI (2026). Statement on multi-agent incident activity. openai.com.
7. Apart Research (2026). Swarm Alignment Hackathon — Sprint Guidelines and Rubric. apartresearch.com/sprints/swarm-alignment-2026.
8. CohereLabs (2025). Aya Expanse 8B. huggingface.co/CohereLabs.